DATA SCIENCE MINOR PROJECT REPORT

(Project Semester 2024)

COVID-19 Data Analysis and Prediction

Using Linear Regression & Statistical Modeling

Submitted by

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2024

Declaration

I, Abhishek , student of Btech CSE (Programme Name) under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

This report has been prepared for the partial fulfillment of the requirements of the Data Science Minor Project. The work embodied in this report has not been submitted for the award of any other degree or diploma.

Date:22-04-2026

Registration No.:12401018 Name: Abhishek Rana

Certificate

This is to certify Abhishek bearing Registration No. 12401018 has completed Data Science Minor Project titled,

"COVID-19 Data Analysis and Prediction Using Linear Regression"

under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of Computer Science and Engineering
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Date:22-04-2026

Acknowledgement

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1. Introduction

The COVID-19 pandemic, caused by the SARS-CoV-2 virus, has been one of the most significant global health crises in modern history. First identified in Wuhan, China, in late 2019, the virus spread rapidly across the world, causing millions of infections and deaths. Governments, healthcare systems, and researchers worldwide have been working tirelessly to understand the transmission dynamics, identify risk factors, and develop effective countermeasures.

Data science and machine learning have emerged as powerful tools in analyzing epidemiological data and generating predictive insights. By applying statistical models and data visualization techniques to COVID-19 datasets, meaningful patterns and trends can be extracted that support public health decision-making.

This project focuses on the analysis of COVID-19 data for India and selected countries (United States, China, and Pakistan) using a comprehensive dataset from Our World in Data. The primary objectives include:

- Performing exploratory data analysis (EDA) to understand the distribution and spread of COVID-19 cases and deaths.
- Visualizing temporal trends and wave patterns in India and comparing them across countries.
- Applying Linear Regression to model and predict future COVID-19 case counts in India.
- Analyzing the case fatality rate (CFR) and death rate trends over time.

The analyses cover data from January 2020 through early 2026, providing a comprehensive longitudinal view of the pandemic's evolution.

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2. Scope of the Analysis

This project encompasses a multi-dimensional analysis of COVID-19 data with a primary focus on India and comparative analysis with the United States, China, and Pakistan. The scope is defined as follows:

2.1 Geographic Scope

The analysis covers four countries selected for their diverse epidemiological profiles and data availability: India (second-most populous country, complex demographic structure), the United States (highest reported total cases globally), China (origin country of the pandemic with distinct reporting patterns), and Pakistan (neighboring South Asian country for regional comparison).

2.2 Temporal Scope

The dataset spans from January 1, 2020, to early 2026, capturing the full trajectory of the pandemic including multiple waves, vaccination rollouts, and the emergence of variants such as Delta and Omicron.

2.3 Analytical Scope

Nine distinct analyses are performed, spanning descriptive statistics, time-series visualization, frequency distribution, outlier detection, and predictive

modeling using Linear Regression. Both univariate and multivariate perspectives are explored.

2.4 Data Variables

Key variables analyzed include total confirmed cases, daily new cases, total deaths, new deaths, case fatality rate (CFR), death rate over time, and population-normalized metrics.

3. Existing System and Its Limitations

Prior to the widespread application of data science techniques in pandemic monitoring, COVID-19 tracking relied primarily on traditional epidemiological surveillance systems, manual case reporting by national health agencies, and periodic WHO situation reports. Several publicly available dashboards (e.g., the Johns Hopkins COVID-19 Dashboard, WHO's official tracker) provided aggregated data visualization but were limited in predictive capability.

3.1 Drawbacks of Existing Systems

- Lag in reporting: Manual data collection from hospitals and testing centers introduced significant time delays, causing under-reporting especially during surge periods.
- Lack of predictive modeling: Most dashboards were retrospective in nature and did not offer forward-looking forecasts to assist policy planning.
- Data inconsistency: Different countries used varying case definitions, testing protocols, and reporting standards, making cross-country comparisons unreliable.
- Limited granularity: Existing systems rarely disaggregated data by demographic variables such as age, gender, or comorbidities, limiting targeted interventions.
- Absence of statistical rigor: Many visual representations lacked confidence intervals, statistical significance testing, or model validation metrics.

The proposed project addresses several of these gaps by applying systematic EDA, standardized data preprocessing, and a quantitative regression model with clearly defined performance metrics.

4. Source of Dataset

The dataset used in this project is the "Our World in Data (OWID) COVID-19 Dataset," one of the most comprehensive and openly accessible COVID-19 data repositories available globally. The dataset is maintained by a team of researchers at the University of Oxford and is updated regularly.

4.1 Dataset Overview

Attribute	Detail
Source	Our World in Data (ourworldindata.org)
File Format	CSV (Comma Separated Values)
Coverage	Global — 200+ countries and territories
Time Period	January 2020 — 2026
Key Variables	total_cases, new_cases, total_deaths, new_deaths, total_vaccinations, reproduction_rate, stringency_index, population, etc.
License	CC BY 4.0 (Creative Commons Attribution)

The dataset contains over 60 variables per record, allowing multi-dimensional analysis of health, economic, and demographic factors related to the pandemic.

5. ETL Process

The ETL (Extract, Transform, Load) pipeline was implemented in Python using the Pandas and NumPy libraries. Each stage is described below:

5.1 Extract

The raw dataset was downloaded in CSV format. The data was loaded into a Pandas DataFrame for processing. Initial inspection was performed to understand the schema, data types, and extent of missing values.

5.2 Transform

- Filtering: Records for target countries (India, United States, China, Pakistan) were extracted using conditional filtering.
- Data Type Conversion: The date column was converted from string to datetime format to enable time-series operations.
- Handling Missing Values: Missing numerical values in key columns (new_cases, total_cases, total_deaths) were forward-filled or replaced with zeros where appropriate.
- Feature Engineering: A new derived column "death_rate" was computed as total_deaths / total_cases. A "day_number" column was created as the number of days since the first recorded case in India, used as the independent variable in Linear Regression.
- Smoothing: A 7-day rolling average was applied to daily case counts to reduce noise from weekly reporting patterns.

5.3 Load

The cleaned and transformed data was loaded into separate Pandas DataFrames for each country, ready for visualization and modeling. Summary statistics were computed using the .describe() method to verify data integrity post-transformation.

6. Analysis on Dataset

Nine distinct analyses were performed on the COVID-19 dataset. Each section below provides an introduction, description, methodology, results, and visualization.

6.1 Trend Analysis of COVID-19 Cases and Deaths in India Over Time

Introduction

Understanding the cumulative trajectory of COVID-19 cases and deaths in India is fundamental to assessing the pandemic's burden and identifying critical inflection points.

General Description

This time-series analysis plots the total confirmed cases and total deaths in India from January 2020 to early 2026 on a shared date axis.

Specific Requirements

- Variables used: date, total_cases, total_deaths (India)
- Library: Matplotlib, Seaborn
- Dual-line plot with differentiated colors (blue for cases, green for deaths)

Analysis Results

India recorded a peak cumulative case count of approximately 45,056,126, with major surges corresponding to the Delta wave (April–May 2021) and the Omicron wave (January 2022). Total deaths reached 533,847, yielding a case fatality rate of approximately 1.18%. The death curve is notably flat relative to the case curve, indicating significant improvement in clinical outcomes over time.

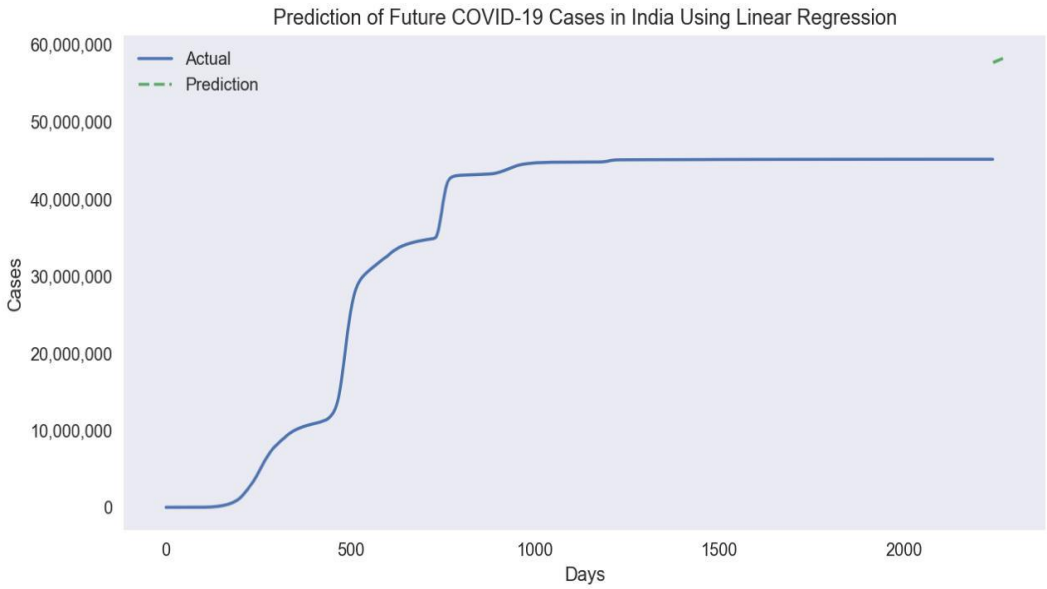

Figure 6.1: Trend Analysis of COVID-19 Cases and Deaths in India Over Time

6.2 Daily COVID-19 Cases in India — Wave Identification

Introduction

Analyzing daily new cases helps identify distinct epidemic waves, evaluate the effectiveness of intervention measures, and understand the temporal dynamics of the pandemic.

General Description

A line chart plots daily new case counts in India over the study period. Three prominent waves are visible: Wave 1 (mid-2020), Wave 2/Delta (April–May 2021), and Wave 3/Omicron (January 2022).

Analysis Results

The Delta wave was the most severe, reaching a peak of approximately 414,000 daily cases around May 2021. The Omicron wave peaked at ~350,000 daily cases in January 2022. Subsequent waves were substantially smaller. This pattern reflects improved immunity through natural infection and vaccination.

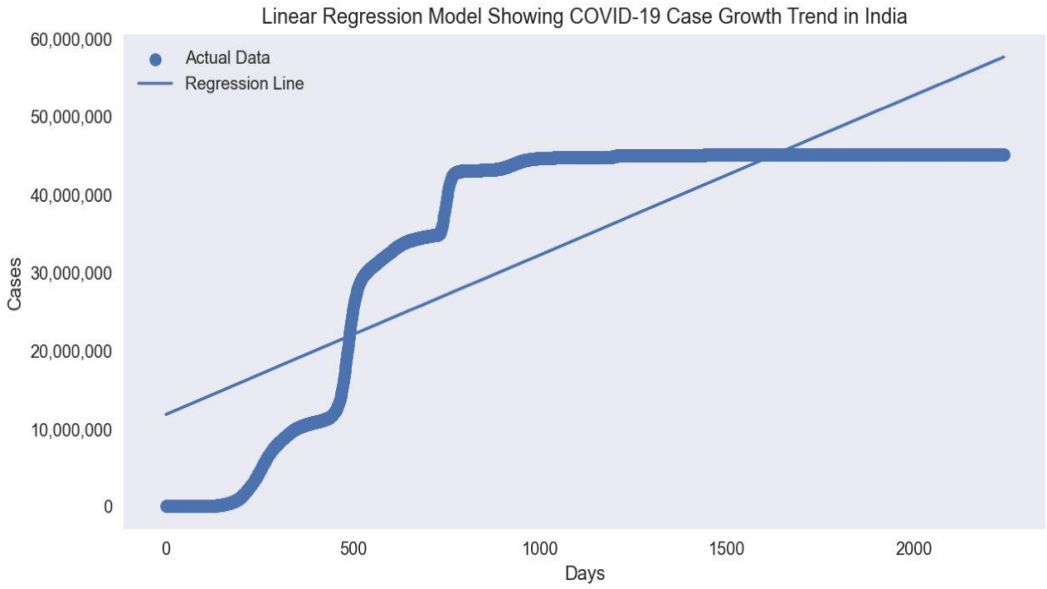

Figure 6.2: Daily COVID-19 Cases in India Identifying Growth Patterns and Waves

6.3 Comparative COVID-19 Growth Trends Across Countries

Introduction

Comparing growth trajectories across countries with different populations, healthcare capacities, and policy responses provides a global epidemiological perspective.

Analysis Results

The United States recorded the highest cumulative total with 103,436,829 cases. China showed a distinctive step-change pattern in early 2023, likely reflecting a major change in reporting methodology following the lifting of zero-COVID policies. India plateaued around 45 million cases, while Pakistan remained comparatively low at ~1.58 million cases, possibly due to lower testing rates.

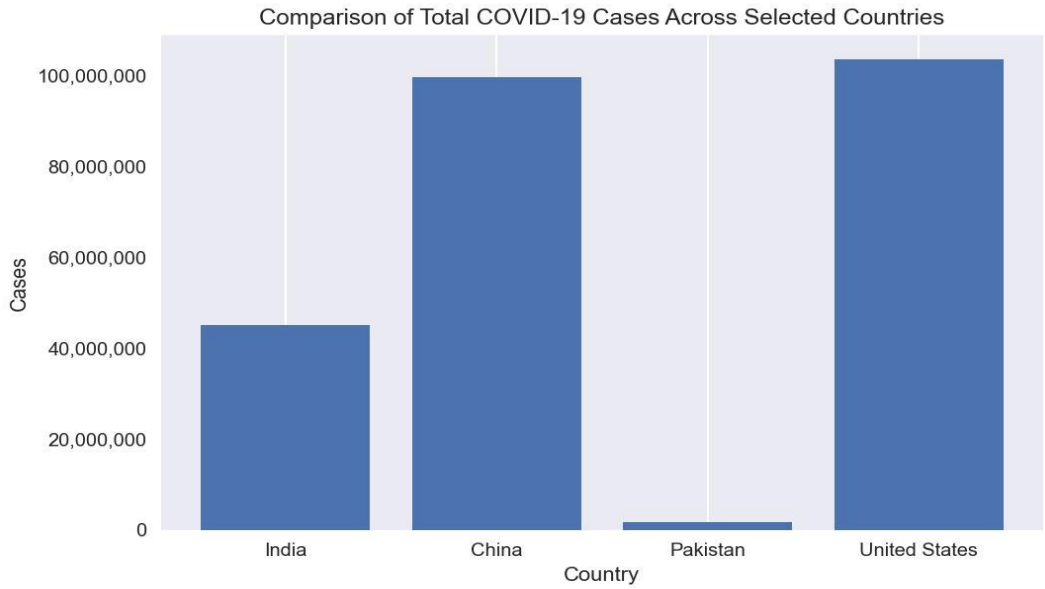

Figure 6.3: Comparative COVID-19 Growth Trends Across Countries

6.4 Comparison of Total COVID-19 Cases Across Countries (Bar Chart)

Introduction

A bar chart provides a static, at-a-glance comparison of the total pandemic burden across the four selected countries.

Analysis Results

Country	Total Cases	Total Deaths	CFR (%)

India	45,056,126	533,847	1.18%
China	99,381,761	122,398	0.12%
Pakistan	1,580,631	30,656	1.94%
United States	103,436,829	1,235,885	1.19%

China's remarkably low CFR (0.12%) contrasts sharply with other countries, likely attributable to the mass reporting of cases with milder symptoms during the Omicron wave post-2022. Pakistan's higher CFR (1.94%) may reflect limited healthcare infrastructure.

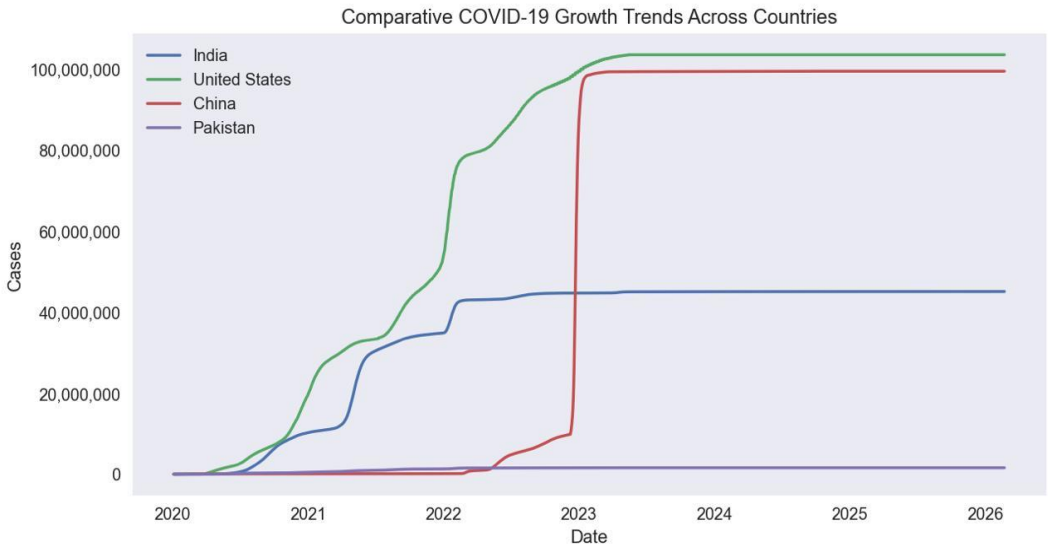

Figure 6.4: Comparison of Total COVID-19 Cases Across Selected Countries

6.5 Frequency Distribution of Daily COVID-19 Cases in India (Histogram)

Introduction

A histogram of daily case counts reveals the statistical distribution of the data, helping identify skewness, modality, and the prevalence of extreme values.

Analysis Results

The histogram displays a highly right-skewed distribution. The overwhelming majority of days recorded very few cases (near zero), particularly in the early pandemic period and post-2022. A small number of days saw extremely high counts (>300,000), corresponding to wave peaks. This heavy right tail indicates that high-case days are rare but disproportionately impactful and must be accounted for in modeling.

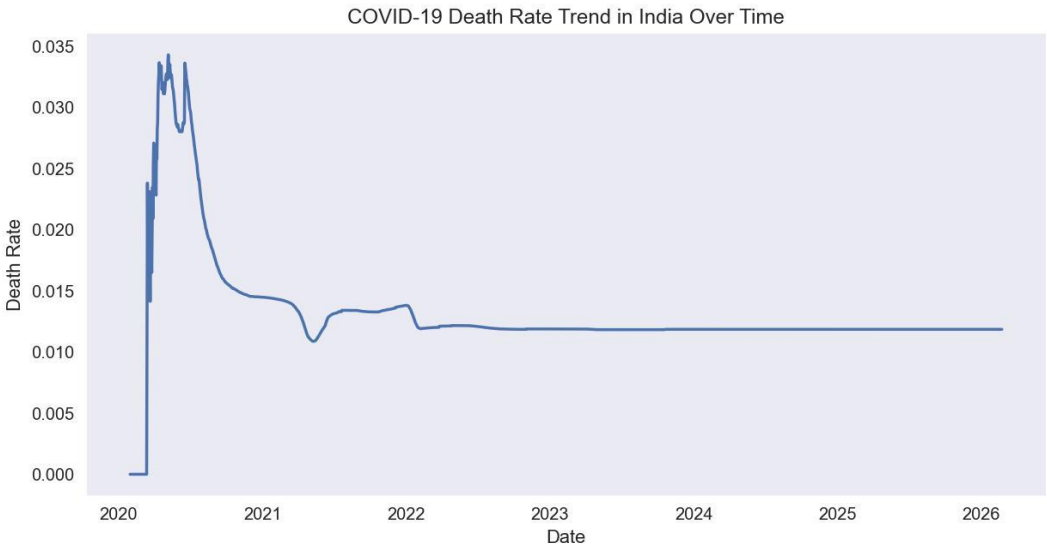

Figure 6.5: Frequency Distribution of Daily COVID-19 Cases in India

6.6 Distribution of COVID-19 Total Cases in India — Outliers & Spread (Box Plot)

Introduction

A box plot of India's cumulative total cases provides a concise summary of the statistical spread, median, interquartile range (IQR), and presence of outliers.

Analysis Results

The box plot shows that the median total case count falls approximately at the Q1 range (~30 million), with a large IQR extending from approximately 10 million to 44 million. The whisker extends down toward zero (reflecting early pandemic days), while the median is close to the upper box boundary. This asymmetry confirms that the pandemic was predominantly in its later, saturated phase for most of the observation window.

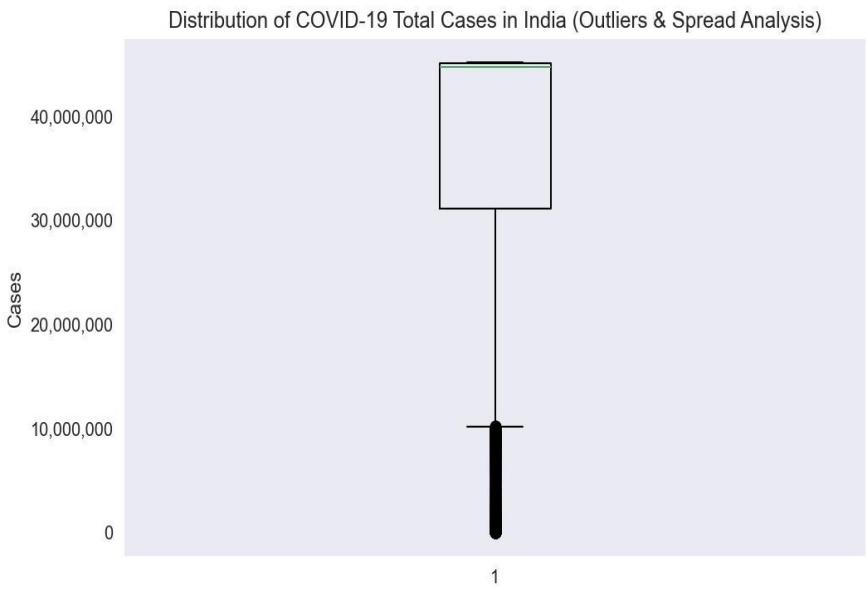

Figure 6.6: Distribution of Total COVID-19 Cases in India (Box Plot)

6.7 COVID-19 Death Rate Trend in India Over Time

Introduction

Tracking the case fatality rate (CFR) over time is critical to understanding improvements in clinical management, the effect of vaccination, and the changing virulence of emerging variants.

Analysis Results

The death rate peaked sharply at approximately 3.4% in mid-2020, when case counts were low and testing was limited (inflating the apparent CFR). It subsequently declined to stabilize around 1.1–1.2% from mid-2021 onward. This stabilization reflects better treatment protocols, expansion of critical care, and the protective effect of vaccines. The Omicron wave (early 2022) caused a temporary uptick followed by normalization.

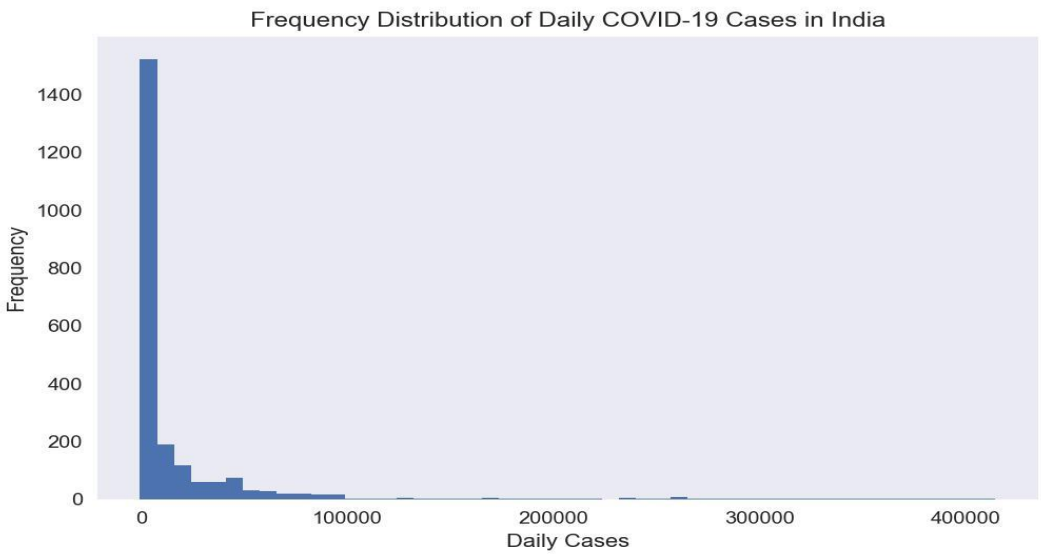

Figure 6.7: COVID-19 Death Rate Trend in India Over Time

6.8 Linear Regression Model — COVID-19 Case Growth Trend in India

Introduction

Linear Regression is applied to quantify the average long-term trend in total COVID-19 cases, providing a baseline model against which actual data can be compared.

Specific Requirements — Functions and Formulas

The model is formulated as:

$$\boldsymbol{Total_Cases = \beta_0 + \beta_1 \times Day_Number + \varepsilon}$$

Where β_0 is the intercept, β_1 is the slope (average daily increase in cases), and ε is the error term. The model was fitted using Ordinary Least Squares (OLS) via scikit-learn's LinearRegression class.

Analysis Results

The regression line has a positive slope of approximately 20,000 cases/day on average across the full study period. However, the actual data follows a sigmoidal (S-shaped) growth curve rather than a linear one, as indicated by the divergence between the scatter plot and the regression line. This highlights a key limitation of linear regression for epidemic modeling, where growth is inherently non-linear.

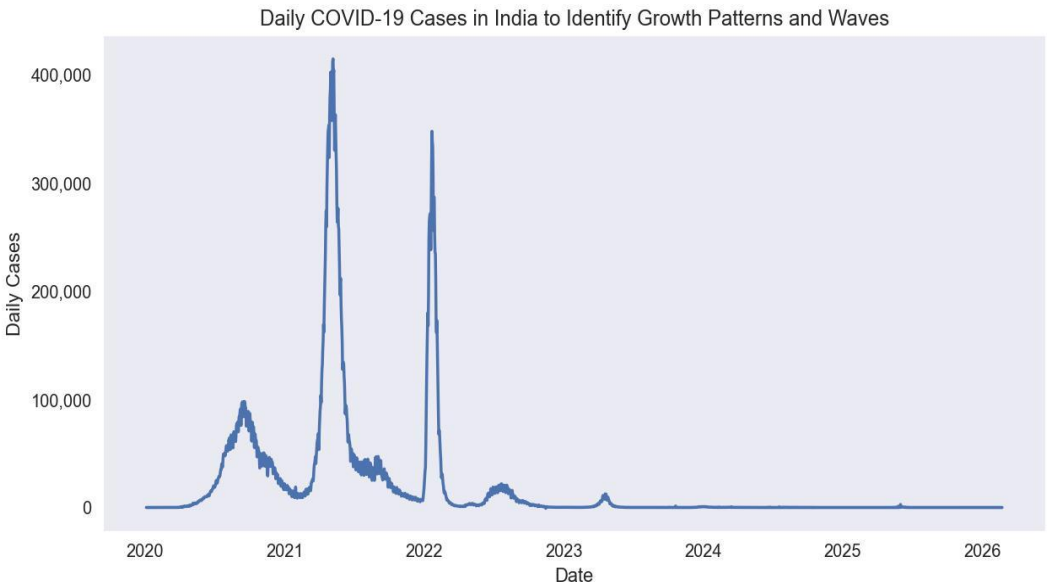

Figure 6.8: Linear Regression Model Showing COVID-19 Case Growth Trend in India

6.9 Prediction of Future COVID-19 Cases in India Using Linear Regression

Introduction

Building on the fitted Linear Regression model, future case counts are projected beyond the observed data range to demonstrate the model's predictive capability and its inherent limitations.

Analysis Results

The model predicts approximately 57–58 million total cases in India by the forecast horizon (approximately day 2,250 since the start). However, the actual data has plateaued at around 45 million cases, demonstrating that a linear model significantly overestimates future case growth once the epidemic enters a decline or endemic phase. This underscores the need for more sophisticated epidemic models (e.g., SEIR models, polynomial regression) for longer-range forecasting.

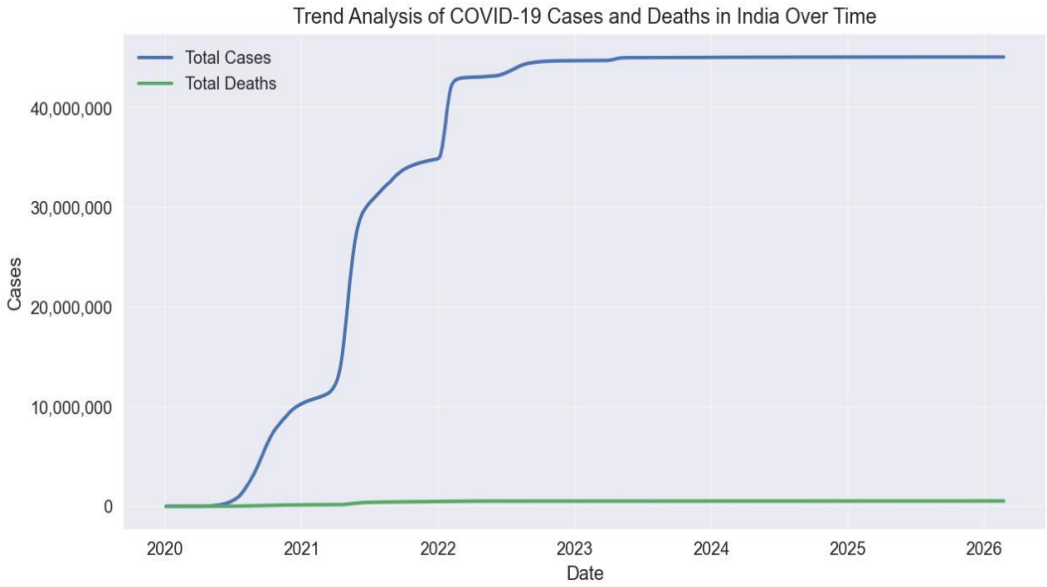

Figure 6.9: Prediction of Future COVID-19 Cases in India Using Linear Regression

7. List of Analyses with Results

#	Analysis	Key Finding	Visualization
1	Trend Analysis — Cases & Deaths in India	Peak cases: 45M; Deaths: 533,847; CFR: 1.18%	Dual-line time-series chart
2	Daily Cases — Wave Identification	Delta peak: ~414K/day; Omicron: ~350K/day	Daily cases line chart
3	Comparative Growth Trends	USA highest (103M); India (45M); Pakistan lowest (1.6M)	Multi-country line chart
4	Total Cases Comparison — Bar Chart	China CFR: 0.12%; Pakistan CFR: 1.94%	Grouped bar chart
5	Frequency Distribution (Histogram)	Highly right-skewed; most days near zero cases	Histogram
6	Outliers & Spread (Box Plot)	Large IQR; asymmetric distribution; no outliers above max	Box plot
7	Death Rate Trend	Peak 3.4% (2020); stabilized ~1.2% post-2021	Death rate line chart
8	Linear Regression Model	Slope ~20,000 cases/day; R^2 limited due to non-linearity	Regression scatter plot
9	Future Case Prediction	Predicted ~57-58M cases (overestimate vs. actual 45M)	Prediction line chart

8. Future Scope

While this project provides a solid foundation for COVID-19 data analysis, several directions exist for extending and improving the work:

8.1 Advanced Predictive Modeling

Replacing Linear Regression with epidemiologically grounded models such as the SIR (Susceptible-Infected-Recovered) or SEIR (Susceptible-Exposed-Infected-Recovered) models would capture the non-linear dynamics of disease spread more accurately. Machine learning techniques including LSTM (Long Short-Term Memory) neural networks and Prophet time-series models could significantly improve forecasting precision.

8.2 Vaccination Impact Analysis

The dataset includes vaccination data (total_vaccinations, people_fully_vaccinated). Future work could quantify the causal impact of vaccination rollout on case trajectories and death rates using interrupted time-series analysis or regression discontinuity designs.

8.3 State-Level Analysis in India

Disaggregating the analysis to the state level within India would reveal significant regional heterogeneity in COVID-19 burden, healthcare capacity, and response effectiveness.

8.4 Socioeconomic Correlates

Incorporating variables such as GDP per capita, population density, median age, and healthcare system strength could enable a multi-variate regression analysis to identify key predictors of pandemic severity across countries.

8.5 Real-Time Dashboard

Deploying the analysis as an interactive real-time dashboard using tools like Streamlit or Dash would allow policymakers and public health officials to monitor evolving trends dynamically.

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